Name: ______________________________ Tutorial group: ________________

Matriculation number: | | | | | | | | | |

For graders only	Question	1	2	3(a)	3(b)	3(c)	4(a)	4(b)	Bonus	Total
	Marks									

November 2024 **MH1812 Test 2** 60 minutes

QUESTION 1. (20 marks)

Solve the linear recurrence relation

$$a_n - 8a_{n-1} + 16a_{n-2} = 0$$

with the initial values $a_0 = 4$ and $a_1 = 16$.

[Proof:] The characteristic equation is $x^2 - 8x + 16 = 0$, which has a double root $x_1 = x_2 = 4$. Therefore,

$$a_n = c_1 4^n + c_2 n 4^n.$$

Using the initial values $a_0 = 4$ and $a_1 = 16$, we have

$$c_1 = 4$$
$$4c_1 + 4c_2 = 16.$$

Thus, $c_1 = 4$ and $c_2 = 0$. It follows that

$$a_n = 4^{n+1}.$$

QUESTION 2. (20 marks)

Use a membership table to prove for arbitrary sets A, B, C that

$$(A \triangle B) - C = (A - C) \triangle (B - C),$$

where $\triangle$ denotes the symmetric difference of two sets, that is, $A \triangle B = (A - B) \cup (B - A)$.

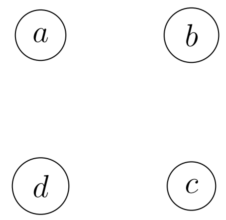
A	B	C	$A \triangle B$	$A \triangle B - C$	$A - C$	$B - C$	$(A-C)\triangle(B-C)$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	1	1	0	1	1
0	1	1	1	0	0	0	0
1	0	0	1	1	1	0	1
1	0	1	1	0	0	0	0
1	1	0	0	0	1	1	0
1	1	1	0	0	0	0	0

The 4th and 7th columns are identical, hence equal.

QUESTION 3. (30 marks)

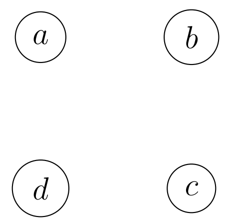
Define a set $A = \{a, b, c, d\}$ and a relation R on A as $R = \{(a, b), (b, c), (c, d), (d, a)\}$.

(a) (10 marks) Find the smallest positive integer i such that R^i is symmetric (justification not required). If such an i exists, draw R^i on the graph provided below, where the vertices are specified, and indicate the edge directions clearly. If such an i does not exist, state "does not exist".



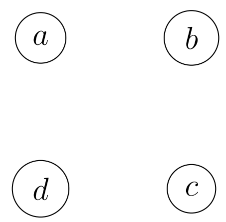
$i = 2$, $R^2 = \{(a, c), (c, a), (b, d), (d, b)\}$.

(b) (10 marks) Find the smallest positive integer j such that R^j is reflexive (justification not required). If such a j exists, draw R^j on the graph provided below, where the vertices are specified, and indicate the edge directions clearly. If such a j does not exist, state "does not exist".



$i = 4$, $R^4 = \{(a, a), (b, b), (c, c), (d, d)\}$.

(c) (10 marks) Draw $R^2 \cup R^4$ on the graph provided below, where the vertices are specified. Indicate the edge directions clearly. Then, determine whether $R^2 \cup R^4$ is an equivalent relation, a partial order or neither.



$R^2 \cup R^4 = \{(a, a), (b, b), (c, c), (d, d), (a, c), (c, a), (b, d), (d, b)\}$, which is reflective, symmetric, transtive, and not anti-symmetric, hence an equivalent relation, not a partial order.

QUESTION 4. **(30 marks)**

How many ways can a team of 4 be selected from a group of 8 people (note that parts (a) and (b) are independent from each other)

(a) (15 marks) if two specific people refuse to be on the same team (i.e., if one is on the team, the other must be excluded)? Your answer should be a specific number, not an unevaluated or partially evaluated expression.

[Solution:]

$$\underbrace{2 \cdot \binom{6}{3}}_{\text{exactly one on the team}} + \underbrace{\binom{6}{4}}_{\text{neither on the team}} = 2 \cdot 20 + 15 = 55.$$

(b) (15 marks) if two specific people must be both on the team together (i.e., if one is on the team, the other must also be included)? Your answer should be a specific number, not an unevaluated or partially evaluated expression.

[Solution:]

$$\underbrace{\binom{6}{2}}_{\substack{\text{the two people are on the team} \\ \text{choose 2 from the remaining 6}}} + \underbrace{\binom{6}{4}}_{\substack{\text{the two people are not on the team} \\ \text{choose 4 from the remaining 6}}} = 2 \cdot \binom{6}{2} = 30.$$

BONUS QUESTION. **(10 marks)**

[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]

How many integer solutions to

$$x_1 + x_2 + x_3 = 2024,$$

are there for which $x_1 \geq 0$, $-1812 \leq x_2 \leq 0$, and $0 \leq x_3 \leq 1812$?

[Solution:] Replacing x_2 with $x_2 + 1812$, we can equivalently find the number of integer solutions to

$$x_1 + x_2 + x_3 = 3836,$$

subject to the constraint $x_1 \geq 0$, $0 \leq x_2, x_3 \leq 1812$.

Now, notice that

$$x_1 = 3836 - x_2 - x_3.$$

For a given pair (x_2, x_3), the solution (x_1, x_2, x_3) is uniquely determined. Valid pairs of (x_2, x_3) include all possibilities in $\{0, \ldots, 1812\} \times \{0, \ldots, 1812\}$. Therefore, the answer is $1813^2 = 3286969$.

[Alternative Solution:] Let $N(n; S)$ denote the number of integer solutions to $x_1 + x_2 + x_3 = n$ subject to constraints $x_1, x_2, x_3 \geq 0$ and the additional constraints in S. As above, the answer we seek is $N(n; x_2, x_3 \leq 1812)$. We have

$$\begin{aligned} N(n; x_2, x_3 \leq 1812) &= N(n; \emptyset) - N(n; x_2 \geq 1813) - N(n; x_3 \geq 1813) + N(n; x_2, x_3 \geq 1813) \\ &= N(n; \emptyset) - N(n - 1813; \emptyset) - N(n - 1813; \emptyset) + N(n - 2 \cdot 1813; \emptyset) \\ &= \binom{n+2}{2} - 2\binom{n - 1811}{2} + \binom{n - 2 \cdot 1813 + 2}{2}. \end{aligned}$$

When $n \geq 2 \cdot 1813$, the expression above equals

$$\begin{aligned} &\frac{1}{2}(n+1)(n+2) - (n - 1811)(n - 1812) + \frac{1}{2}(n - 3624)(n - 3625) \\ &= 1 - 1811 \cdot 1812 + \frac{3624 \cdot 3625}{2} \\ &= 1 + (3625 - 1811)1812 \\ &= 1 + 1814 \cdot 1812 \\ &= 1813^2 (= 3286969). \end{aligned}$$